A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

Hüseyin Tayyer Canseven

Department of Electrical Engineering, LUT School of Energy Systems, LUT University, Lappeenranta, 53850, Finland

Abstract

Detection methods for incipient stator-winding faults are developed one machine at a time, so how the generator type changes the problem is not quantified, and the detectability measures of fault-detection theory presuppose a model or a trained detector. A statistic that requires neither is proposed, the signal-to-drift ratio (SDR): the relative change of a feature between a healthy reference interval and a faulted interval, divided by a quantile of its changes between adjacent healthy intervals, the known-onset mean-shift test as a standardised effect size calibrated on each machine's own healthy non-stationarity. From it follow a minimum detectable extent, a detectable share and a signature location per machine type, sensor suite and fault class; a decision table selects the cheapest suite that meets a requirement with a stated power and bootstrap confidence; and a transfer test fixes the feature-referencing rule under which a detector serves several machine types. On open laboratory-bench data from permanent-magnet, squirrel-cage induction and wound-field synchronous generators with the same tapped short-circuits, the same design-level fault is three different physical events, phasor residuals remove a faulted-phase dependence of the magnitude ratios, the signature relocates between terminal voltage, controller states and field current with the machine–drive combination, the cheapest adequate suite depends on machine type and requirement, and a learned detector transfers once its features are referenced to the target machine's healthy behaviour. Code and data are

Email address: huseyin.canseven@lut.fi (Hüseyin Tayyer Canseven)

open.

Keywords: condition monitoring, fault detectability, stator winding fault, wind-turbine generators, sensor suite selection, transfer across machines

1. Introduction

Stator-winding insulation faults are the archetypal incipient failure of an electrical machine: a short-circuit across a few turns whose subtle signature evolves into a machine-destroying fault if it is not detected [1]. Electrical-subsystem failures contribute substantially to wind-turbine downtime [2], and a rich literature detects winding faults in a given machine [3–9], almost all of it validated on one machine. Yet the generator type of a large turbine—permanent-magnet synchronous (PMSG), squirrel-cage induction (SCIG), wound-field synchronous (WFSG) or doubly-fed induction—is chosen on cost, mass, efficiency and grid-compliance grounds [10], and its condition monitoring is added afterwards from commercial systems [11, 12]. Whether the same fault is equally visible in different generator types, in which measured quantity it appears, and which sensors are therefore worth specifying for each has not been quantified.

Such questions need a measure of detectability that requires neither a model of the machine nor a trained detector, is comparable across features, operating points and machines, and connects to the quantities a monitoring specification is written in: the smallest fault to be detected, the false-alarm rate tolerated and the sensors it costs. The detectability notions of fault detection and isolation (FDI) theory [13–15] and the sensor-placement methods built on them [16] presuppose a model; a learned detector [17] presupposes data from the very machine, and its transfer across machines is an open problem [18].

This paper proposes the signal-to-drift ratio (SDR): the relative change of an observable feature between a healthy reference interval and a faulted interval, divided by a quantile of the relative changes the same feature exhibits between adjacent healthy intervals when nothing is wrong. It is the known-onset mean-shift test of change detection [19] as a standardised effect size. From it follow a *minimum detectable extent* (MDE), a *detectable share* (DS) above a required extent and a *signature location* per machine type, sensor suite and fault class, a decision table that selects the cheapest adequate suite, and a transfer test that fixes the feature-referencing rule under

which a detector built on one machine type serves another. The question is made answerable by open benchmark datasets [20–22] in which a PMSG, an SCIG and a variable-speed WFSG were subjected to the same inter-turn and inter-winding short-circuits at the same tapped positions of the stator winding, the PMSG and SCIG on one bench with one converter, controller and protocol.

The contributions are the SDR with its screen, calibrated oracle and derived quantities, evaluated on the physically distinct fault sizes of a test matrix rather than on its nominal extents; a procedure that turns them into a decision table with bootstrap confidence and a transfer test over referencing rules; and the first quantitative comparison known to the author of winding-fault detectability across three generator types, with four findings: the design-level extent and the physical severity of a fault are different axes, and the faulted phase rides on the nominal extent axis unless sequence quantities are taken as phasor residuals; the measurement group in which a fault appears depends on the machine–drive combination; the cheapest adequate suite depends on the machine type *and* on the requirement; and a learned detector transfers once its features are referenced to healthy behaviour of the target machine.

2. Background and related work

Fault detection and isolation has a mature theory of diagnosability for modelled systems: structural analysis determines which faults are detectable and isolable from which sensor sets [14, 15], sensor placement selects the cheapest adequate set [16], and Basseville [13] defines detectability as a distance between the residual distributions under the healthy and the faulted hypothesis; all presuppose a trusted model. Data-driven diagnosis [17] replaces the model by data from the very machine, with transfer across machines an active problem [18], and prognostics grades features by trendability [23] or by separability measures such as the per-feature area under the receiver operating characteristic curve (AUC) [24]. The SDR belongs to that family, with the sensitivity index d', the fault-to-noise ratio of model-based FDI and the per-feature AUC; what is new is its null, the drift of each feature between adjacent healthy intervals of every trial, which fixes the false-alarm rate of the simplest detector and makes the magnitude comparable across features, operating points and machines, together with the screen, the limit-of-detection quantities and their use to compare machine types and sensor

suites.

Inter-turn short-circuits are among the most studied incipient faults in electrical machines [1, 25]. Negative-sequence methods [4, 5, 26], the extended Park's vector approach [27] and the reviews [6, 7, 28] have established the signatures: a winding asymmetry produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous frame, and because the fault-induced negative-sequence phasor adds to the inherent asymmetry of the machine, the detection quantity is the change of that phasor rather than of its magnitude [4, 26]. Under closed-loop current control the asymmetry migrates into the voltage commands and controller states to a degree set by the current-loop bandwidth relative to twice the electrical frequency [8, 9, 29]; in a wound-field machine the stator negative sequence induces a component at twice the electrical frequency in the field current [30], and in permanent-magnet machines with an accessible neutral the zero-sequence voltage is a sensitive indicator [31, 32]. These facts have not been turned into a measure applicable across candidate machine types; learned detectors are developed for one machine [33], and finite-element turn-fault models [34] are the natural trial generators before hardware exists.

3. Method

3.1. Definitions and terms

Let $\mathcal{A}$ be a set of candidate *alternatives*, each a machine type with the control structure it implies; $\mathcal{F}$ a set of failure-mode classes, each parameterised by a design-level *extent* $e \in E_f$ realisable in every alternative; $\mathcal{S}$ a set of *sensor suites*, each a set of measurements, including quantities internal to the drive controller, with a cost $c(s)$ and the set $\Phi(s)$ of features computable from them; and $\mathcal{O}$ the operating envelope, discretised into operating points o. Because the design-level extent does not fix the physical severity of the event (Section 5.1), a *physical severity axis* is declared beside it. A *trial* is a record of alternative a at operating point o with a healthy reference interval R immediately followed by a faulted interval F in which (f, e) is present; trials may come from a physical test or from a validated simulation (Section 3.5).

3.2. The signal-to-drift ratio

The reference interval is split into two adjacent halves R_1, R_2, with R_2 adjacent to F. For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_{R_1}, \bar{\varphi}_{R_2}, \bar{\varphi}_F$ be its

mean values over the intervals and define the relative change under fault, referenced to the half adjacent to the fault, and the relative change between the two healthy halves,

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_{R_2}}{|\bar{\varphi}_{R_2}|}, \qquad \delta^0_\varphi = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}. \tag{1}$$

δ^0_φ measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. It is available from every trial of alternative a, faulted or not, and its absolute value pooled over the N_a trials forms the empirical null of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\mathrm{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha\big(|\delta^0_\varphi|\big)}, \tag{2}$$

where q_α is the α-quantile of the pooled null; with $\alpha = 0.95$, $\mathrm{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable interval pairs. For sequence quantities, whose fault-induced phasor adds to the inherent asymmetry of the machine, φ is the complex phase-referenced negative-sequence ratio and the changes are magnitudes of complex differences (Section 4.3).

The SDR is a mean-shift statistic with a known change time, studentised by an empirical quantile of its own drift null: for a zero-mean Gaussian null $q_{0.95}(|\delta^0|) = 1.96\,\sigma_{\delta^0}$ and $\mathrm{SDR} \geq 1$ is a two-sided 5 % test on the shift of interval means, the fixed-sample, known-onset case of the generalised likelihood ratio of Basseville and Nikiforov [19]; the quantile is kept because the null has an excess kurtosis of up to 7 here, and its sampling variability (a factor 1.3 to 1.5 at 95 %) enters every interval reported. With a known onset, a whole faulted interval averaged and a reference immediately before the fault, the SDR characterises the simplest detector on the feature, an abrupt-onset threshold at a stationary operating point against a null of sub-second non-stationarity; a detector that combines features or accumulates evidence can detect below $\mathrm{SDR} = 1$, which is why the calibrated oracle and the transfer experiment are part of the method. Because a quieter null raises the SDR as much as a stronger signature does, numerator and denominator are reported separately when alternatives are compared (Section 5.2).

Safeguards. (i) On healthy trials, in which the "faulted" interval contains only the switching operation, a feature should exceed $\mathrm{SDR} = 1$ at the nominal

rate $1-\alpha$; a feature whose false-alarm count in the n_h healthy trials is incompatible with that rate (one-sided binomial test at the 5 % level; three or more of nine at $\alpha = 0.95$) or whose null quantile exceeds one is excluded for that alternative. With nine healthy trials the screen has little power (a true rate of 20 % passes with probability 0.74), so every false-alarm count is reported. (ii) A suite is evaluated by one *fixed* feature, chosen once among its screened features by the smallest minimum detectable extent, then the detectable share, then the median SDR, or by the *oracle* feature, the best feature of each trial, at a calibrated threshold, the α-quantile of $\max_\varphi |\delta^0_\varphi| / q_\alpha(|\delta^0_\varphi|)$ over all records, because a maximum over k features exceeds the single-feature threshold at a rate of up to $1 - \alpha^k$. (iii) Every design quantity carries a bootstrap interval, over tap pairs for statements about a class and over operating points within an extent level for the MDE, with the null quantile re-estimated in every resample.

3.3. Derived quantities

For each alternative a, suite s and class f: the *diagnosability map* $D_a(s, f, e, o)$ is the SDR of the suite's fixed feature at extent e and operating point o; the *minimum detectable extent* $\mathrm{MDE}_a(s, f)$ is the smallest tested extent level e^* such that $\mathrm{median}_o\, D_a(s, f, e, o) \geq 1$ for every tested $e \geq e^*$, reported as the bracket $(e^-, e^*]$ between the last undetected and the first detected level, the levels being the physically distinct fault sizes of the test matrix (Section 4.2); the *detectable share* $\mathrm{DS}_a(s, f \mid e \geq e_{\mathrm{req}})$ is the fraction of faulted trials with $e \geq e_{\mathrm{req}}$ in which SDR ≥ 1, pooled over the tested levels at or above e_{req}, so that it depends on the composition of the test matrix; the share at the tested level that e_{req} maps to is the stricter, operator-facing form, and Section 5.5 reports the verdicts under both; and the *signature location* is the measurement group—stator currents, terminal voltages, controller internals, mechanical quantities, excitation—whose best screened feature attains the highest SDR in the largest share of trials. For a requirement $(f, e_{\mathrm{req}}, \alpha)$ the *adequate* suites of an alternative are those with $\mathrm{MDE} \leq e_{\mathrm{req}}$ and a detectable share above the requirement of at least a stated power; the signature location tells the designer *why* an alternative behaves as it does.

3.4. Procedure

The procedure (Fig. 1) has six steps; steps 3–4 are fixed by Eqs. (1)–(2), the others are design decisions with default rules.

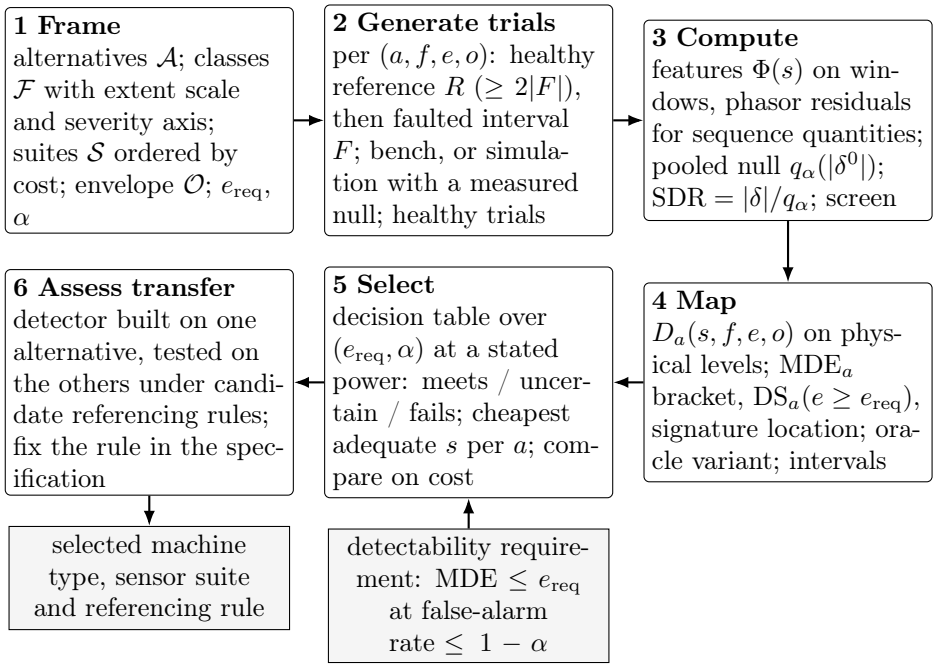


Figure 1: The six-step procedure; steps 3–4 are fixed by Eqs. (1)–(2), steps 1, 2, 5 and 6 follow the default rules of Section 3.4

1. **Frame.** Enumerate $\mathcal{A}$, the classes $\mathcal{F}$ with an extent scale realisable in every alternative and a physical severity axis, the suites $\mathcal{S}$ ordered by cost and the envelope $\mathcal{O}$; set e_{req} as the smallest extent whose growth to an unacceptable consequence outpaces the inspection interval and $1 - \alpha$ as the per-trial false-alarm rate tolerated.
2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference of at least twice the faulted duration immediately followed by the fault, covering at least the levels that bracket e_{req} and the corners of the envelope, plus healthy trials with the same timing for the screen.
3. **Compute.** Extract $\Phi(s)$ on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, apply the screen.
4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into MDE_a, DS_a and the signature location for the fixed and oracle variants, with bootstrap intervals.
5. **Select.** Build the decision table: for each alternative, class, suite and candidate (e_{req}, α), classify the suite as *meets* when the bootstrap prob-

ability of the joint event $\{\mathrm{MDE} \le e_{\mathrm{req}}$ and $\mathrm{DS}(e \ge e_{\mathrm{req}}) \ge \mathrm{DS}_{\mathrm{req}}\}$ is at least 0.95, *fails* when it is at most 0.05 and *uncertain* otherwise, $\mathrm{DS}_{\mathrm{req}}$ being the power the requirement demands (0.9 here); the fixed feature is re-selected and the null re-estimated in every resample, the operating-point trials of a level are resampled with its tap pairs treated as exchangeable (the tap-pair unit is reported where the two differ), a suite's verdict is at least that of its subsets, and e_{req} takes the tested levels, because between levels the MDE is only a bracket. The adequate suite is the cheapest that meets the requirement for every class; the choice of machine type is affected only when no affordable suite meets it for some alternative.

6. **Assess transfer.** If the detector must serve a family of alternatives, test it across them under candidate feature-referencing rules; the rule that transfers enters the specification.

3.5. Design-stage instantiation and scope

Before hardware exists the two ingredients of the SDR are supplied separately: the numerator by a finite-element or circuit model of each alternative with the fault injected [34], validated on a few physical trials, the denominator from measured healthy records of hardware of the same class, on the testable assumption that the healthy non-stationarity depends on the sensing and control chain more than on the machine. The method applies to classes that can be seeded at a known extent, at a stationary operating point, with a healthy interval immediately before the fault, in every alternative; it evaluates a threshold on interval means, not the time to detection; and $1-\alpha$ is a per-decision rate, at one decision per interval (0.36 s here) a false alarm every 7 s for $\alpha = 0.95$, so the MDEs reported are bench limits of detection, not field thresholds, and a field monitor needs an accumulation rule over many decisions.

4. Demonstration on three wind-turbine generators

4.1. Alternatives and data

The three alternatives are laboratory-scale generators on instrumented test benches at the University of São Paulo, published as open benchmark datasets (Table 1, Fig. 2). The PMSG [20] and the SCIG [21] are four-pole, 230 V, double-star machines, rated 2.5 kVA and 2.5 kW, on the *same* bench: the same prime mover, converter, control platform and field-oriented control

(FOC; rotor angle from an encoder for the PMSG, from a flux observer for the SCIG), sensors, acquisition (20 kHz, 3 s records), fault-insertion hardware and test script, so that everything but the machine and the control structure it requires is held fixed. The variable-speed WFSG [22] is a 2 kVA four-salient-pole machine on an earlier bench of the same group [35] with a three-level converter, FOC with encoder angle, 4 kHz acquisition and 1.3 s records: the same tapped short-circuits on a different bench, whose fault resistances, fault durations, converter, sampling rate and voltage sensing all differ and whose variable-speed dataset holds no healthy records, so that the screen and the nuisance-transient negatives are unavailable for it. The data papers state neither the loop gains nor the switching frequencies, observer structure or transducer bandwidths.

Table 1: The three alternatives; the PMSG and SCIG share one bench and one protocol, the WFSG bench differs where indicated

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, static exciter
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.3 s
Reference / faulted interval	0.68 s / 0.36 s	0.68 s / 0.36 s	0.49 s / 0.06–0.28 s
Fault resistance	2.6 Ω	2.6 Ω	1, 2.83, 5.66, 11.32 Ω
Operating points	3 speeds × 3 torques	same	716–1910 rpm, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials	216	216	438 (198 at 2.83 Ω)

Ratings as given by the manufacturer (kVA for the synchronous machines, kW for the induction machine). 2L: two-level converter; 3L-NPC: three-level neutral-point-clamped converter; FOC: field-oriented control. Current-loop gains and bandwidths, switching frequencies and the flux-observer structure are not given in the data papers.

4.2. Failure-mode class and extent scale

All three machines have 24 tapped derivation points at the same relative positions of their stator windings. Two classes are used (Fig. 3): inter-turn short-circuits (both taps on one branch) and inter-winding short-circuits (taps on the two parallel branches of a phase); the extent e is the fraction of a branch between the taps, 2.7–12.1 % for the twelve inter-turn and 2.3–40.2 % for the twelve inter-winding cases. The nominal extents overstate the

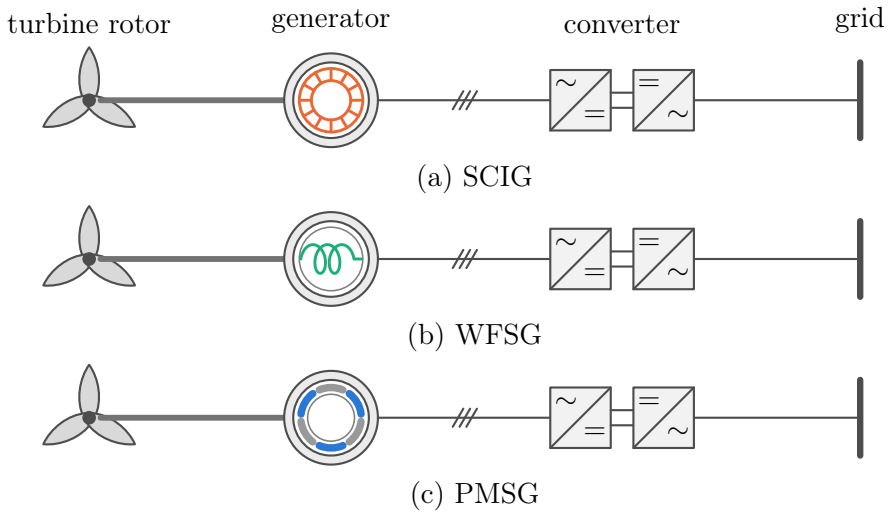
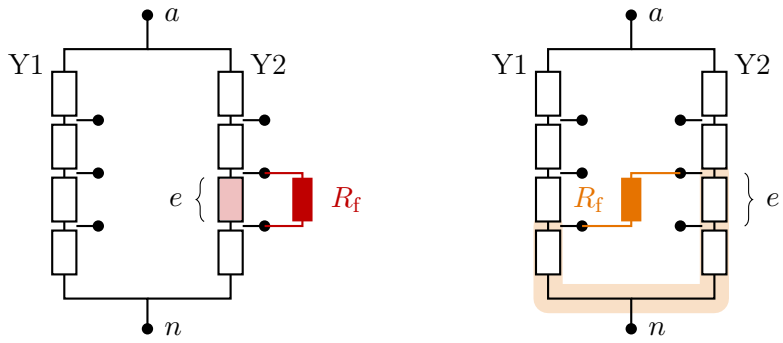


Figure 2: The three alternatives as generator systems: turbine rotor, generator with its rotor excitation (cage, field winding or magnets), converter and grid

resolution of the matrix: the fault current of a resistor-limited short-circuit is set by the voltage induced in the shorted turns, so tap pairs with equal fault currents short the same number of turns and are the same physical fault (on the PMSG the four pairs of each inter-turn level carry fault currents within 3 % of each other, 1.4, 3.8 and 6.0 A; the SCIG currents cluster within 8 %). The limit-of-detection quantities are therefore evaluated on *physical extent levels*, tap pairs of one class whose median fault currents, normalised per machine and averaged over the two machines, lie within 7 % of the first pair of the level: three inter-turn levels (2.7–3.8, 7.4–7.7 and 11.6–12.1 % of a branch) and six inter-winding levels (2.3–3.2, 9.7–10.6, 14.3–14.8, 21.7–22.2, 33.7–37.5 and 40.2 %). The tap layout has a second property that matters for sequence features: the four smallest inter-turn pairs lie in phase A, the 7.4–7.7 % pairs in phases A and B and the largest in phases A and C, so on the nominal extent axis the faulted phase is confounded with the extent (Section 5.3). Faults were closed through a contactor and a series resistance at steady state, 2.6 Ω for 0.4 s on the PMSG and SCIG and 1 to 11.32 Ω on the WFSG, where the resistance was raised with extent, so that extent and impedance are confounded on that bench; the severity is *resistor-limited*, unlike a bolted fault, which the leakage reactance of the shorted turns limits.

The PMSG and SCIG matrix is complete: 24 fault cases × 3 speeds (1200,



(a) Inter-turn short-circuit (b) Inter-winding short-circuit

Figure 3: Inter-turn (**a**) and inter-winding (**b**) short-circuits through R_f between taps of one phase of the double-star stator; e is the fraction of a branch between the taps

1500, 1800 rpm) × 3 torques (5.2, 6.4, 8.0 N m, 39–60 % of rated torque), scaled from the operating region of the International Energy Agency 15 MW reference turbine [36] by the speed ratio, plus the healthy condition at each operating point; the SCIG generates with slip, at 59.6 Hz at 1800 rpm. The WFSG matrix covers the same 24 tap pairs at six speeds (716–1910 rpm) and four torque levels, sparsely. The analyses use the WFSG's 438 inter-turn and inter-winding trials at all impedances for the null and the feature ranking, with the impedance as a covariate, and the 198 trials at 2.83 Ω, the impedance closest to the PMSG/SCIG value, covering four inter-turn pairs of 11.6–12.1 % and four inter-winding pairs of 2.3–10.6 %, for the MDE and the decision table.

4.3. Sensor suites and features

Five sensor suites are considered, ordered by the data access, hardware and integration they add to a bare generator–converter set (Table 2): *drive-internal* uses only quantities the converter controller already has; *3 CT* uses the three phase currents with angle-free features, so that its cost is data access rather than hardware; *3 CT + 3 VT* adds three terminal voltage transducers (VT); *3 CT + 3 VT + drive* adds controller access; and *+ torque transducer* adds a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite), available because its exciter is static [30]. Sequence quantities enter in two forms: the magnitude ratios I_2/I_1, V_2/V_1 and D_2/D_1 of currents, terminal voltages and duty cycles, the established indicators [4, 5, 27]; and the phasor residuals

$|\Delta\bar{I}_2|/I_1$, $|\Delta\bar{V}_2|/V_1$ and $|\Delta\bar{D}_2|/D_1$, the magnitude of the change of the complex negative-sequence component referenced to the positive-sequence angle, which isolates the fault-induced part whatever the faulted phase [4, 26]: with X_+ and X_- the symmetrical components of the Hann-windowed complex amplitudes of the three phase signals at f_e, the phase-referenced ratio of a window is $\bar{X}_2 = X_-\overline{X_+}/|X_+|^2$, whose angle does not rotate with the window start; the interval mean of a residual feature is the complex mean of $\bar{X}_2$ over the windows of the interval, and its relative changes are $|\bar{X}_{2,F} - \bar{X}_{2,R_2}|/|\bar{X}_{2,R_2}|$ under fault and $|\bar{X}_{2,R_2} - \bar{X}_{2,R_1}|/|\bar{X}_{2,R_1}|$ in the null. Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle; the electrical frequency and the phase rotation are estimated per window from the slope of the unwrapped phase of the current space vector, so that the external suites need no angle sensor; the other features are the standard deviation of the mean-removed signal and its amplitudes at f_e and $2f_e$ as listed in Table 2; on the WFSG bench the converter-side phase voltages serve as terminal voltages and the electrical torque is computed from power and speed.

Table 2: Sensor suites and the features they can compute

Suite	Added access / hardware	Features
Drive-internal	controller data	I_d, I_q: std, $1f_e$, $2f_e$; $V_d^{\mathrm{cmd}}, V_q^{\mathrm{cmd}}$: $2f_e$; PI outputs d, q: std, $2f_e$; duty-cycle negative sequence D_2/D_1; T_e: std, $1f_e$, $2f_e$; DC-link std; speed std
3 CT	three phase currents, no angle	I_2/I_1, RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q: $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m: std, $1f_e$, $2f_e$
Excitation (WFSG)	field-current measurement	I_f: std, $2f_e$

f_e: electrical frequency; $1f_e$, $2f_e$: amplitude of the component at once and twice the electrical frequency; I_1, I_2 (V_1, V_2): positive- and negative-sequence magnitudes at f_e; std: standard deviation; THD: total harmonic distortion; PI: proportional–integral.

4.4. Computation

For the PMSG and SCIG the reference interval is the 0.68 s before the contactor command, split into R_1 (0.32 s) and R_2 (0.36 s), and the faulted

interval runs from 30 ms after the command to 10 ms before its release, so that, the relay picking up 85–90 ms later, it holds about 55 ms of healthy time, which makes the PMSG/SCIG SDRs slightly conservative; for the WFSG the fault interval starts where the 10 ms RMS of the measured fault current first exceeds $\max(0.3\,\mathrm{A}, 6\sigma)$ and the reference interval (0.49 s) is split at 0.30 s, four trials with no complete fault window being dropped. A window belongs to R_1 when it ends before the split time and to R_2 when it starts after it; the reference intervals fall 6 % (common bench) and up to 13 % (longest WFSG faults) short of the twice-the-fault-duration rule of step 2. The main PMSG–SCIG comparison uses $n_c = 5$; the three-alternative comparison uses $n_c = 3$ because the WFSG fault intervals are short; the sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and $\alpha \in \{0.90, 0.95, 0.99\}$ is reported. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 WFSG records. Bootstrap intervals use 2000 resamples: the median SDR and the detectable share of a class are resampled over the twelve tap pairs, the same pairs for both machines so that the SCIG/PMSG ratio is matched, the MDE over the operating-point trials within each extent level, and in every resample the null quantile is re-estimated from a cluster resample of the machine's 25 clusters; the decision table uses 1000 resamples per suite, alternative, class and α, with e_{req} at the tested levels and a required detectable share of 0.9.

Step 6 is instantiated as a window-level change detector: a gradient-boosted decision-tree (GBDT) classifier [37] and, for the PMSG–SCIG pair, a logistic regression, trained to separate faulted windows from healthy ones, within an alternative by five-fold cross-validation grouped by tap pair and across alternatives by training on all trials of one machine and testing on all trials of another. Four feature-referencing rules are compared: physical units; features z-scored on all healthy trials of the target machine (a commissioning baseline); z-scored on the healthy trial at the same operating point; and the absolute z-score of each window against the *first half* of its own trial's reference interval. Negatives are the reference and recovery windows of all trials plus the fault-interval windows of the healthy trials (141 and 140 of about 21 000 out-of-reference windows on the PMSG and SCIG), the only negatives that contain the contactor operation without a fault; because the normalising windows are trivially separable under the fourth rule, every metric is reported on the out-of-reference windows, with 95 % intervals from 500 cluster resamples of the tested machine's trials. The GBDT is a histogram gradient boosting of depth 3 with 200 iterations at a learning rate of 0.08,

the logistic regression is standardised with $C = 1$, the within-machine folds are grouped by tap pair (healthy trials by speed), and the normalising set of the self-referenced rule is the windows ending before 0.62 s (PMSG/SCIG) or 0.30 s (WFSG).

5. Results

5.1. The same design-level fault is not the same physical fault

For the same tap pair the SCIG's fault current is about 10 % lower than the PMSG's (medians 3.3 against 3.7 A for inter-turn and 7.7 against 8.5 A for inter-winding pairs), but relative to what the stator sensors see the difference is larger: the fault current is 1.45 times the pre-fault stator current (RMS, median over pairs) in the PMSG and 0.63 times in the SCIG for inter-turn faults, and 3.3 against 1.5 times for inter-winding faults, because the SCIG's stator current carries the magnetising component (5.3 against 2.6 A RMS before the fault); the WFSG pairs at the matching impedance give 4.7 and 3.1, that bench running at a stator current of 1.1 A. A fault specified as "e % of a winding" therefore arrives at the sensors of the three alternatives as three different physical events. The physical severity axis, the RMS fault current relative to the RMS stator current, orders faults of equal turns correctly because the MMF asymmetry of a turn fault scales with the fault current times the shorted fraction; across levels it is a measured, not a general, severity scale.

5.2. Where the signature appears, and why the SDR differs

The null of the SDR differs between the PMSG and the SCIG on every feature, and by how much depends on the form in which the feature is written: in the relative form of Eq. (1) the SCIG's null is 2.4 times quieter than the PMSG's for the voltage residual, 5.2 times for the current residual and 45 times for the d-axis current ripple; in the units of the features themselves the factors are 1.1, 6.0 and 7.0, because the SCIG's healthy d-axis ripple is five times the PMSG's: its inherent negative-sequence current is 2.9 times the PMSG's in amperes (0.043 against 0.015 A), and the flux-observer angle, modulated at $2f_e$, is the plausible remainder. The decomposition therefore reads differently for the three measurement groups: through the terminal-voltage residual the machines are equivalent, numerator and null agreeing in volts to within 20 %; through the stator-current residual the SCIG's SDR is 3.7 and 3.3 times the PMSG's while the PMSG's current change is the larger

in both forms, so there the SCIG wins entirely by a quieter null; through the d-axis features the SCIG's SDR is 16 and 23 times the PMSG's and in absolute units the SCIG has both the larger signature and the quieter null. The reading that the SCIG wins only by silence thus holds for the stator current and is, for the controller quantities, an artefact of the relative normalisation; nor is the null a property of the transducers alone, since the duty-cycle residual, a controller output, has a null 8.5 (relative) and 3.1 (absolute) times larger on the PMSG.

Fig. 4 shows the median SDR of every feature for the three alternatives on three-cycle windows. In the PMSG the terminal-voltage group carries the strongest signature in 94 % of inter-turn and 97 % of inter-winding trials, through the voltage residual (median SDR 6.8 and 53 at five-cycle windows; 88 and 100 % of trials detectable), and the stator-current residual is weak for inter-turn faults (1.7; 70 %). In the SCIG the controller group is strongest in 63 and 67 % of trials, through the duty-cycle residual for inter-turn faults (11.2; 93 %) and the d-axis ripple and its PI output for inter-winding faults (68; 100 %), the stator-current group in 34 and 33 %; In the WFSG the terminal-voltage and current residuals, the commanded q-axis voltage and the field-current $2f_e$ ripple, an observable the other two lack, are strongest. The physical quantity behind the relocation is the incremental negative-sequence impedance at the terminals, $\Delta\bar{V}_2/\Delta\bar{I}_2$ per trial: 12.7 [11.1, 14.1] Ω (inter-turn) and 12.2 [10.2, 13.2] Ω (inter-winding) in the PMSG against 7.3 [6.9, 7.7] and 7.4 [7.2, 7.6] Ω in the SCIG 0.60 and 0.35 per unit on the 2.5 kVA, 230 V base, so the PMSG drive converts a negative-sequence current disturbance into 1.7 times the terminal voltage that the SCIG drive does. With the converter connected directly to the stator, this ratio is the closed-loop output impedance of the current regulator at the negative-sequence frequency [8, 29], tied to the machine through the inductance the regulator is tuned to; how much of the difference is tuning cannot be checked without the undocumented gains. The $2f_e$ ripple of the synchronous-frame currents is the same negative-sequence current seen in the rotating frame, so the current and controller groups differ by normalisation and angle source rather than by physics; both machines run the same field-oriented controller but differ in angle source, output impedance and healthy baselines, so the relocation belongs to the machine–drive combination, not to the control structure alone.

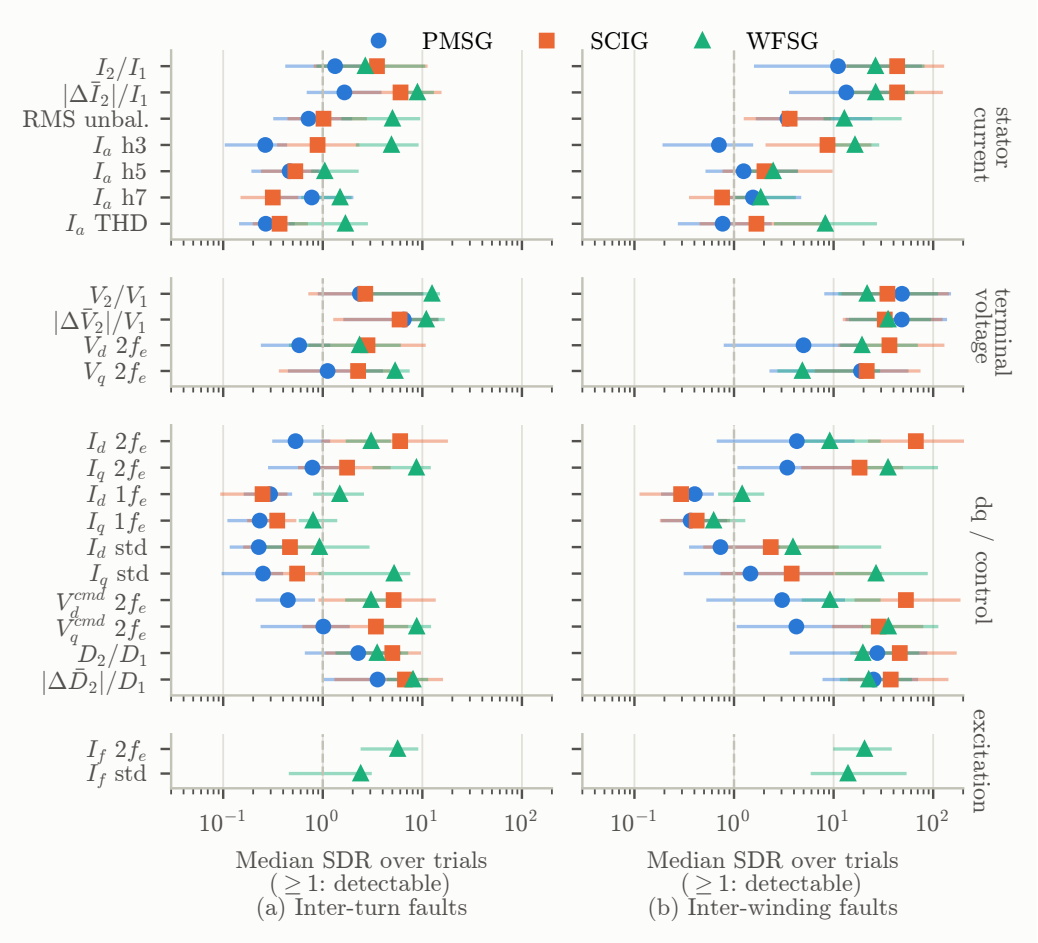


Figure 4: Median SDR of every feature (dot) with interquartile range (bar) for the three alternatives, three-cycle windows; the WFSG is pooled over its fault impedances

5.3. The faulted phase and the extent axis

The inter-turn test matrix carries a second variable on its nominal extent axis: the four pairs of the smallest level lie in phase A, those of the middle level in phases A and B, those of the largest in phases A and C. Fig. 5 shows why this matters: the magnitude ratios V_2/V_1 and I_2/I_1 depend on the faulted phase far beyond what the extents explain (the spread between the pairs of one level is a factor 3.8 to 14 for V_2/V_1 and 1.3 to 7 for I_2/I_1), whereas the phasor residuals remove this almost entirely, the pairs of every level agreeing to within a factor 1.1 to 1.8 on both machines. The mechanism is the one

of Section 4.3: the inherent negative-sequence component of each machine is of the same order as the fault-induced one for a small inter-turn fault, so the magnitude of their sum rises or falls with the angle between them, which rotates by ±120° with the faulted phase; the complex change does not. Hence the residual features represent the suites wherever a sequence quantity is used (Tables 3–5), the magnitude ratios remaining in Fig. 4 as the established indicators; and on the residual features the two machines on the common bench are nearly indistinguishable through the terminal voltage, with per-level medians of $|\Delta\bar{V}_2|/V_1$ of 1.1, 6.8 and 19.5 (PMSG) against 0.9, 6.9 and 19.5 (SCIG) at the three inter-turn levels, and differ through the stator current, where the SCIG's residual is 2 to 4 times the PMSG's: the relocation of the signature survives the change of feature, the earlier reading that the PMSG's voltage signature is stronger than the SCIG's does not.

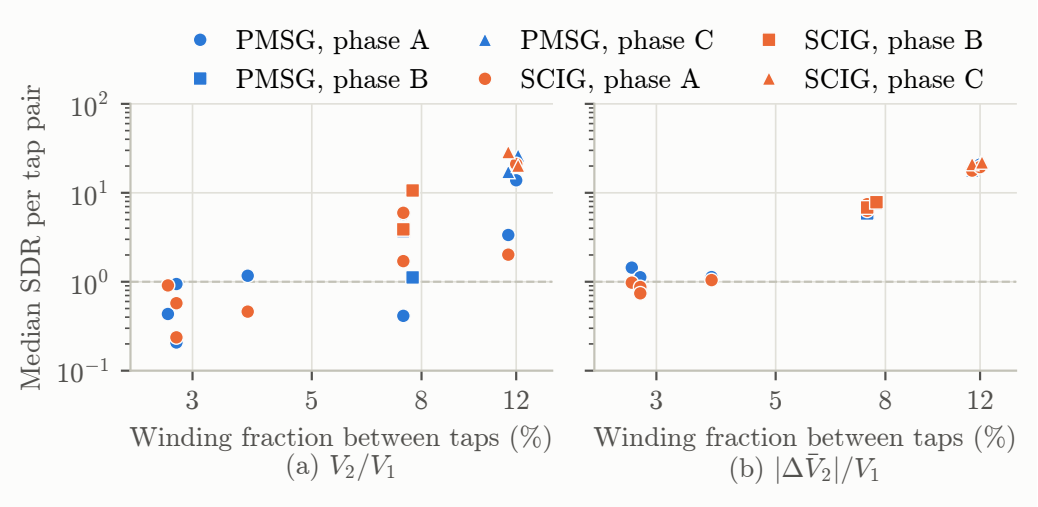


Figure 5: Inter-turn median SDR per tap pair against the nominal extent: magnitude ratio V_2/V_1 (**a**) and phasor residual $|\Delta\bar{V}_2|/V_1$ (**b**), five-cycle windows; marker, faulted phase; colour, machine

5.4. Minimum detectable extent

Table 3 gives, for the two residual features, the d-axis PI ripple and the q-axis voltage ripple, the MDE as a bracket on the physical levels, the detectable share and the median SDR with intervals from the tap-pair bootstrap, and the SCIG/PMSG ratio of medians. Through the terminal-voltage

residual the PMSG detects every inter-turn level including the smallest (per-level median SDR 1.1; pooled share 88 % [74, 98], 64 % at that level) and the SCIG every level from the middle one (its smallest level sits at a median of 0.9); through the stator-current residual the point estimates reverse, while the operating-point intervals of both MDEs span the two smallest levels; the d-axis ripple detects from the middle level on the SCIG and only the largest on the PMSG. For inter-winding faults the PMSG detects from the smallest level (2.3–3.2 %) through the voltage residual, with every trial detected, but only from 9.7–10.6 % through the current residual and from 14.3–14.8 % through the d-axis ripple, while the SCIG detects from the smallest level on every feature except the q-axis voltage ripple. The SCIG/PMSG ratios of median SDR confirm the equivalence through the voltage residual (1.0 [0.7, 1.3] for inter-turn faults) and the SCIG's advantage through the current residual (3.7 [2.1, 5.1]) and the d-axis ripple (17 [3, 42]); on the physical severity axis the SCIG's residuals lie an order of magnitude above the PMSG's at equal fault-to-stator current ratio, because its ratio is formed with a stator current that carries the magnetising component, whereas at equal fault current in amperes the two machines' voltage residuals nearly coincide and the SCIG's current residual stays 2 to 5 times the PMSG's; the WFSG, at a ratio of 3 to 5 at the matching impedance, is comparable on neither axis without its impedance as a covariate.

5.5. Sensor suites and the selection

Table 4 gives the suite results for the PMSG and SCIG at five-cycle windows. For the PMSG, current transducers alone detect inter-turn faults from the middle level and inter-winding faults from 9.7–10.6 %; three voltage transducers take both to the smallest level through the voltage residual (88 and 100 % of trials); the drive-internal suite reaches the same inter-turn bracket as the current transducers through the duty-cycle residual, and the smallest inter-winding level (96 % of trials). For the SCIG the cheapest suites already reach what any suite reaches: the smallest level of both classes, through the current residual or, with no added sensor, the duty-cycle residual (93 % of inter-turn trials) and the d-axis PI output (100 % of inter-winding trials); at three cycles every suite stops at the middle level for inter-turn faults. At three cycles the WFSG detects all inter-winding faults from 2.3 % with every suite and 92 % of the inter-turn trials at the matching impedance, the field current alone 100 and 80 %. The fixed features have at most two false alarms in nine healthy trials (Wilson interval 6–55 %), a nested selection of the fixed

Table 3: Minimum detectable extent, detectable share and median SDR with 95 % intervals, four features, PMSG and SCIG, five-cycle windows

Class	Feature	M.	MDE [CI]	DS [CI]	Median SDR [CI]	SCIG/PMSG [CI]
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	≤3.0 [3.0–7.5]	88 [74, 98]	6.84 [1.67, 15.9]	
		S	(3.0, 7.5] [3.0–7.5]	81 [63, 95]	6.9 [1.16, 16.4]	1.01 [0.683, 1.29]
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	(3.0, 7.5] [7.5]	70 [44, 93]	1.7 [0.751, 4.14]	
		S	≤3.0 [3.0–7.5]	85 [70, 96]	6.34 [1.93, 13.2]	3.72 [2.08, 5.08]
IT	$\mathrm{PI}_d\,2f_e$	P	(7.5, 11.8] [11.8]	24 [7, 44]	0.37 [0.289, 0.797]	
		S	(3.0, 7.5] [3.0–7.5]	76 [57, 93]	6.18 [1.22, 18.8]	16.7 [3.04, 41.7]
IT	$V_q\,2f_e$	P	(7.5, 11.8] [7.5–11.8]	51 [31, 73]	1.13 [0.649, 3.96]	
		S	(3.0, 7.5] [7.5]	60 [34, 84]	2.34 [0.47, 5.07]	2.08 [0.492, 3.35]
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	≤2.8 [2.8]	100 [100, 100]	52.7 [15.3, 148]	
		S	≤2.8 [2.8]	100 [99, 100]	39 [14.9, 136]	0.74 [0.607, 1.1]
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	(2.8, 10.1] [10.1]	86 [68, 100]	13.7 [4.1, 48.3]	
		S	≤2.8 [2.8]	99 [95, 100]	45.8 [16.5, 104]	3.33 [2.08, 6.64]
IW	$\mathrm{PI}_d\,2f_e$	P	(10.1, 14.6] [10.1–21.9]	69 [49, 88]	3 [0.937, 10.1]	
		S	≤2.8 [2.8]	100 [100, 100]	68.4 [23.6, 223]	22.8 [14.2, 36.7]
IW	$V_q\,2f_e$	P	(2.8, 10.1] [2.8–10.1]	89 [77, 98]	18.7 [4.23, 55.1]	
		S	(2.8, 10.1] [2.8–10.1]	88 [71, 100]	22.5 [7.26, 68.4]	1.2 [0.76, 1.97]

$\alpha = 0.95$. MDE on the physical extent levels (% of a branch): $(a, b]$, level a not detected and every level from b up detected; $\leq b$, every tested level detected; in brackets its 95 % interval over the operating-point trials of each level. DS (%), median SDR and the SCIG/PMSG ratio of medians carry percentile intervals from a bootstrap over tap pairs (the same pairs for both machines) with the null quantile re-estimated in every resample. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

feature on the other tap pairs changes no share by more than six points, and the calibrated oracle exceeds the fixed feature's share in one suite only (by three points) and otherwise equals it or falls below it by up to 22 points, so combining the features of a suite buys nothing over its best feature once the false-alarm rate is held. A shaft torque transducer adds nothing for winding faults (median SDR below 0.3 on the PMSG, unbounded null on the SCIG).

The decision table (Table 5) turns this into the selection of step 5. For inter-turn faults at the smallest level (2.7–3.8 % of a branch) no suite on either machine meets the requirement at $\alpha = 0.95$: the PMSG's voltage suite (probability 0.27) and the SCIG's drive-internal suite (0.82) are uncertain, the joint event failing in 73 and 18 % of the resamples while the fixed features detect 64 and 78 % of the trials at that level itself, and the current-only suites fail. From the middle level (7.4–7.7 %) every suite on both machines meets it with probability 1.00; on the WFSG only the largest level is testable at the matching impedance, and there every suite meets it except the field current alone (0.92). For inter-winding faults the SCIG and the WFSG meet the re-

Table 4: Sensor suites versus what they detect, PMSG and SCIG, five-cycle windows: fixed feature, MDE bracket, detectable share and false alarms

Suite	PMSG				SCIG			
	feature	MDE	DS (oracle)	FA	feature	MDE	DS (oracle)	FA
Inter-turn faults								
drive	$\lvert\Delta\bar{D}_2\rvert/D_1$	(3.0, 7.5]	75 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (71)	2/9
3CT	$\lvert\Delta\bar{I}_2\rvert/I_1$	(3.0, 7.5]	70 (61)	0/9	$\lvert\Delta\bar{I}_2\rvert/I_1$	≤3.0	85 (78)	1/9
3CT+3VT	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (75)	0/9	$\lvert\Delta\bar{I}_2\rvert/I_1$	≤3.0	85 (77)	1/9
3CT+3VT+drv	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (74)	2/9
+torque	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (74)	2/9
Inter-winding faults								
drive	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤2.8	96 (90)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9
3CT	$\lvert\Delta\bar{I}_2\rvert/I_1$	(2.8, 10.1]	86 (89)	0/9	I_2/I_1	≤2.8	99 (98)	0/9
3CT+3VT	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (98)	0/9	V_2/V_1	≤2.8	100 (99)	0/9
3CT+3VT+drv	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (98)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9
+torque	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (96)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9

drive: drive-internal quantities alone; 3CT: three current transducers; +3VT, +drv, +torque: voltage transducers, controller access and a shaft torque transducer added cumulatively to 3CT. DS (%) of the fixed feature, in parentheses the share of the calibrated oracle (Section 3.2); FA: false alarms of the fixed feature on the nine healthy trials; MDE as in Table 3; screened-out features are excluded.

quirement at the smallest level with every suite including the drive-internal one under the pooled power, whereas the PMSG needs voltage transducers there (drive-internal 0.93 with 78 % of the trials at that level detected, current-only fails) and meets it with every suite from 9.7–10.6 %. Relaxing α to 0.90 admits the SCIG's drive-internal suite at the smallest inter-turn level and the PMSG's at the smallest inter-winding level; tightening it to 0.99 turns the uncertain inter-turn verdicts into failures and the PMSG's current-only verdict at the middle level into an uncertain one (0.53); on three-cycle windows the SCIG's inter-turn verdicts at the smallest level fail for every suite (Section 5.7).

Table 6 lists the seven cells whose verdict depends on the definition of the power or on the resampling unit; every other cell of Table 5, the WFSG's included, keeps its verdict under all four variants. With the power required at the level itself, the four uncertain cells fail (the fixed features detect 64 and 78 % of the trials at those levels) and two cells that meet under the pooled power become uncertain: the SCIG's current transducers at the smallest inter-winding level (0.77, with 94 % of that level's trials detected) and the PMSG's drive-internal suite at 9.7–10.6 % (0.88, every trial detected, the null

Table 5: Decision table at $\alpha = 0.95$: bootstrap probability that a suite meets the requirement at each tested extent level

Suite	PMSG				SCIG				WFSG			
	probability of meeting the requirement at e_{req} (%)											
Inter-turn faults												
e_{req} (%)	3.0	7.5	11.8		3.0	7.5	11.8		3.0	7.5	11.8	
drive	0.00	**1.00**	**1.00**		0.82	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT	0.00	**1.00**	**1.00**		0.047	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT+3VT	0.27	**1.00**	**1.00**		0.047[b]	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT+3VT+drv	0.27[b]	**1.00**	**1.00**		0.83	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
exc	–	–	–		–	–	–		n.t.	n.t.	0.92	
Inter-winding faults												
e_{req} (%)	2.8	10.1	14.6	21.9	2.8	10.1	14.6	21.9	2.8	10.1	14.6	21.9
drive	0.93	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT	0.00	**0.97**	**0.97**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT+3VT	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT+3VT+drv	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
exc	–	–	–	–	–	–	–	–	**1.00**	**1.00**	n.t.	n.t.

Meets: MDE $\leq e_{\text{req}}$ and detectable share ≥ 0.9 for $e \geq e_{\text{req}}$, e_{req} at the physical levels of the common bench (% of a branch; the WFSG, unscreened for lack of healthy trials, is assessed against the same values on its own tested extents); 1000 resamples of the operating-point trials within each level (the tap pairs of a level treated as exchangeable), null quantile and fixed feature re-estimated in every resample. Bold: probability ≥ 0.95 (meets); below 0.05 the suite fails; n.t.: no tested extent on the required side of e_{req}. PMSG and SCIG at five-cycle windows, WFSG at three cycles and its matching fault impedance; the inter-winding levels 35.9 and 40.2 % are omitted (every suite meets them). Suites as in Table 4; exc: WFSG field current.

[b] Probability inherited from a cheaper subset of the suite (a suite may always use the features of its subsets).

and the fixed feature re-estimated). Resampling tap pairs instead of trials moves the SCIG's drive-internal inter-turn probability from 0.82 to 0.89 and the PMSG's voltage suite from 0.27 to 0.23 and changes no verdict.

5.6. Transfer across alternatives

Table 7 reports step 6 for the PMSG–SCIG pair on five-cycle windows: the GBDT detector on the out-of-reference windows, with 95 % cluster-bootstrap intervals and the true-positive rate (TPR) at 1 % false-positive rate. In physical units the detector reaches an AUC of 0.93 and 0.91 within a machine on unseen tap pairs but 0.79 [0.68, 0.88] (PMSG→SCIG) and 0.61 [0.50, 0.72] (SCIG→PMSG) across, with a TPR of 0.25 and 0.26 across against 0.74 and 0.67 within. Any referencing to healthy behaviour of the target machine recovers most of the AUC, the self-referenced rule reaching 0.93 [0.87, 0.98] and

Table 6: Decision-table cells whose verdict depends on the definition of the power or on the resampling unit within a level

Machine	Class	Suite	e_{req} (%)	DS at level	pooled		at level	
					trials	pairs	trials	pairs
PMSG	inter-turn	3CT+3VT	3.0	64	0.27	0.23	0.00	0.01
PMSG	inter-turn	3CT+3VT+drv	3.0	64	0.27	0.23	0.00	0.01
PMSG	inter-winding	drive	2.8	78	0.93	0.93	0.12	0.00
PMSG	inter-winding	drive	10.1	100	1.00	1.00	0.88	0.94
SCIG	inter-turn	drive	3.0	78	0.82	0.89	0.04	0.00
SCIG	inter-turn	3CT+3VT+drv	3.0	78	0.83	0.92	0.049	0.00
SCIG	inter-winding	3CT	2.8	94	1.00	1.00	0.77	0.71

$\alpha = 0.95$; every other cell of Table 5 keeps its verdict under all four variants. DS at level: share of the fixed feature at e_{req} itself (%). Pooled: share over all tested levels $\geq e_{\mathrm{req}}$ (Table 5); at level: share at e_{req}; trials, pairs: resampling unit within a level. Meets at ≥ 0.95, fails at ≤ 0.05.

0.89 [0.82, 0.95] against 0.87–0.88 for the two baselines; the rules separate on the TPR, where the self-referenced rule gives 0.81 [0.67, 0.94] and 0.54 [0.37, 0.69] against 0.50–0.71 and 0.54–0.62 for the baselines, so that it is the best on AUC in both directions and on TPR in one, with overlapping intervals in the other, and on the linear detector, a logistic regression transferring at 0.93 and 0.91 when self-referenced but at 0.49–0.81 under the baselines. Between the three alternatives, on three-cycle windows, self-referenced GBDT detectors transfer in all six directions with an AUC of 0.85 [0.79, 0.91] to 0.98 [0.96, 1.00] and a TPR of 0.41 to 0.94, against 0.90–0.97 and 0.71–0.93 within a machine, and a detector trained on two machines detects the faults of the third with an AUC of 0.87 (PMSG), 0.90 (SCIG) and 0.99 (WFSG; optimistic, as that bench has no healthy trials).

5.7. Operating envelope, robustness and the two nulls

Table 8 gives the operating envelope. Speed dominates: between 1200 and 1800 rpm the median inter-turn SDR of the fixed features of Table 4 rises by a factor of 1.3 to 4.4 and the detectable shares from 69–86 to 83–100 %, consistent with a resistor-limited fault current that scales with speed; torque changes the medians at a fixed speed by less than a factor 1.25 on the SCIG and up to 1.6 on the PMSG; and the PMSG's current residual, the feature of its retrofit suite, is the exception, falling from 1.8 to 1.4 with speed with its share fixed at 67 %. The terminal impedance $|\Delta\bar{V}_2/\Delta\bar{I}_2|$ does not explain that exception: it is flat within 5 % over $2f_e = 80$ to 120 Hz

Table 7: Transfer of a GBDT detector between PMSG and SCIG under four referencing rules, out-of-reference windows, 95 % intervals

Direction	Metric	physical units	z, all healthy trials of the target	z, healthy trial at the operating point	$\lvert z\rvert$ against own reference
P→P	AUC	0.93 [0.86, 0.98]	0.93 [0.86, 0.98]	0.91 [0.84, 0.96]	0.94 [0.90, 0.98]
	TPR	0.74 [0.55, 0.89]	0.74 [0.55, 0.89]	0.63 [0.43, 0.83]	0.78 [0.64, 0.91]
S→S	AUC	0.91 [0.83, 0.97]	0.91 [0.83, 0.97]	0.91 [0.84, 0.96]	0.96 [0.92, 0.99]
	TPR	0.67 [0.50, 0.85]	0.67 [0.50, 0.85]	0.62 [0.43, 0.79]	0.82 [0.68, 0.94]
P→S	AUC	0.79 [0.68, 0.88]	0.88 [0.81, 0.95]	0.87 [0.79, 0.95]	0.93 [0.87, 0.98]
	TPR	0.25 [0.14, 0.35]	0.50 [0.36, 0.67]	0.71 [0.57, 0.87]	0.81 [0.67, 0.94]
S→P	AUC	0.61 [0.50, 0.72]	0.88 [0.81, 0.94]	0.88 [0.80, 0.94]	0.89 [0.82, 0.95]
	TPR	0.26 [0.10, 0.40]	0.54 [0.32, 0.71]	0.62 [0.42, 0.76]	0.54 [0.37, 0.69]

AUC and true-positive rate at 1 % false-positive rate (TPR) on the out-of-reference windows (Section 4.4); intervals from 500 cluster resamples of the tested machine's trials. P→P, S→S: cross-validation grouped by tap pair; P→S, S→P: trained on one machine, tested on the other.

on both machines (13.0 to 12.5 Ω on the PMSG, 7.5 to 7.2 Ω on the SCIG), so it is not proportional to frequency as a leakage reactance would be, and the bandwidth mechanism of Section 6.2 is not visible inside this band; the speed dependence of the SDRs comes from the fault current and the null, and the PMSG's falling current residual is a drive effect that these data do not resolve. Because the pooled null is a mixture over operating points, the 5 % false-alarm reading holds on average over the envelope: the share of healthy changes above the pooled 95th percentile is 0 to 16 % by speed for the key features, the largest being the SCIG's current residual at 1800 rpm, where its 59.6 Hz electrical frequency lies next to the 60 Hz grid frequency, a candidate cause. Both make the lowest monitored speed a parameter of the requirement.

With the screen applied at every setting, the MDE brackets of the key features are unchanged across window lengths of three, five and eight cycles on both machines except at the margin (the SCIG's current residual reaches the smallest inter-turn level at five and eight cycles but not at three; the PMSG's q-axis voltage ripple reaches the middle level only at eight cycles; the PMSG's d-axis features, whose null quantile exceeds one at eight cycles, are screened out there), and the detectable shares move by at most 7 points except for the PMSG's d-axis features, which lie below SDR = 1 at every setting. Tightening α from 0.95 to 0.99 lowers the shares by 0 to 14 points, moves the MDE up one level for the PMSG's voltage residual and the SCIG's current residual and screens out the PMSG's d-axis features; relaxing it to

Table 8: Operating envelope: fixed inter-turn features and terminal impedance by speed, PMSG and SCIG, five-cycle windows

Machine	Feature / class	1200 rpm	1500 rpm	1800 rpm	Torque factor
Median SDR (detectable share, %) of the fixed inter-turn features					
PMSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	5.3 (78)	7.0 (89)	10.0 (97)	1.36
PMSG	$\lvert\Delta\bar{D}_2\rvert/D_1$	2.7 (69)	4.0 (72)	5.6 (83)	1.64
PMSG	$\lvert\Delta\bar{I}_2\rvert/I_1$	1.8 (67)	2.6 (78)	1.4 (67)	1.59
SCIG	$\lvert\Delta\bar{I}_2\rvert/I_1$	3.8 (69)	6.3 (86)	16.6 (100)	1.21
SCIG	$\lvert\Delta\bar{D}_2\rvert/D_1$	9.5 (86)	11.6 (94)	12.2 (97)	1.18
SCIG	$\lvert\Delta\bar{V}_2\rvert/V_1$	5.8 (78)	7.3 (83)	8.2 (83)	1.15
Incremental negative-sequence impedance $\lvert\Delta\bar{V}_2/\Delta\bar{I}_2\rvert$ *(Ω), median [IQR]*					
PMSG	inter-turn	13.0 [11.5, 14.4]	12.6 [11.3, 14.1]	12.5 [10.0, 13.8]	
PMSG	inter-winding	12.6 [11.1, 13.3]	12.1 [9.2, 13.7]	11.9 [9.8, 12.8]	
SCIG	inter-turn	7.5 [7.0, 8.0]	7.2 [6.7, 7.4]	7.2 [6.9, 7.4]	
SCIG	inter-winding	7.5 [7.4, 7.8]	7.4 [7.3, 7.6]	7.3 [7.1, 7.4]	

Speeds 1200, 1500 and 1800 rpm ($2f_e$ = 80, 100 and 120 Hz; 79, 99 and 119 Hz on the SCIG, which generates with slip); each cell pools the three torques (36 inter-turn trials per machine and speed). Torque factor: largest ratio of per-torque medians at a fixed speed. The impedance is the per-trial complex ratio of the phasor residuals (Section 5.2), 36 trials per cell.

0.90 raises the shares by up to 10 points and moves four MDEs down one level; the orderings of the two machines are the same at every setting. At $\alpha = 0.95$ the screen removes only the SCIG's duty-cycle magnitude ratio (three false alarms in nine healthy trials) and the torque features.

Replacing the reference immediately before the fault by the healthy trial recorded at the same operating point at another time, a commissioning baseline, widens the null of the four key features by a factor of 1.1 to 4.5 (1.9 to 3.2 for the voltage and current residuals), lowers the median SDR of the inter-turn residuals from 6.8 to 2.4 (PMSG, voltage) and from 6.3 to 2.1 (SCIG, current) and their detectable shares from 88 and 85 % to 72 and 69 %, while inter-winding faults remain detectable in 67–93 % of trials.

6. Discussion

6.1. What the demonstration establishes about the method

The demonstration supports four claims about the method and corrects the naive form of three of them. *The design-level description of a failure mode does not fix its physical severity, and the test matrix can carry a hidden variable*: the same winding fraction produced fault currents differing by a factor of 2.3 relative to the stator current between the PMSG and the

SCIG, and the faulted phase rode on the nominal extent axis, which the phasor residuals remove and the magnitude ratios do not; a comparison of alternatives must carry both axes and use features invariant to the nuisance variables of the matrix. *Where a failure becomes observable is a property of the machine–drive combination, to be read with the decomposition in both forms*: "the control structure decides" is not supported, since both machines run the same controller and are indistinguishable through the voltage residual, and "the SCIG wins only by a quieter null" holds for the stator current but not for the controller quantities; a design team must know whether an alternative wins by signature or by silence, because a signature belongs to the fault physics and a null to the sensing and control chain, which a design-stage instantiation cannot simulate (Section 3.5). *The cheapest adequate sensor suite depends on the alternative and on the requirement, and here it changes the monitoring specification, not the choice of machine type*: at the middle inter-turn level the PMSG and the SCIG meet the requirement with drive-internal quantities, and the WFSG does so at the one level testable at its matching impedance; at the smallest inter-turn level no suite on the common bench meets it with the stated power, and the WFSG was not tested there; for inter-winding faults at the smallest level the PMSG alone needs added sensors. *A monitoring design transfers across alternatives under a healthy-referenced rule, which belongs in the specification*: every healthy-referenced rule recovered most of the AUC lost in physical units and the self-referenced rule the most, in both directions and across all three machines, the rules differing in the true-positive rate and for the linear detector; the stronger claim that a detector transfers *only* when self-referenced is not supported. A specification should name the referencing rule and the horizon over which the reference is renewed, since the self-referenced rule is an abrupt-onset detector that a slowly growing fault would evade.

6.2. Implications for the condition monitoring of wind-turbine generators

Within the scope of Section 6.3, a PMSG with full-power conversion should be specified with terminal voltage measurement, or with another observable of the terminal asymmetry such as the zero-sequence voltage where a neutral is accessible [31], if inter-winding faults of the smallest tested size (2.3–3.2 % of a branch) are to be detected by threshold monitoring with the stated power; for inter-turn faults the voltage measurement raises the share of detected trials at the smallest level from 11–25 to 64 % (pooled over all levels, from 70–75 to 88 %) without making that level a confident requirement, and

from the middle level the controller's duty-cycle residual suffices. A third-party retrofit limited to current and voltage transducers reads Table 5 as follows: on the PMSG, three current transducers meet the requirement from the middle inter-turn level and from 9.7–10.6 % for inter-winding faults, voltage transducers are needed at the smallest inter-winding level, and nothing meets the smallest inter-turn level; on the SCIG, three current transducers meet every inter-winding level under the pooled power (the smallest level is uncertain at the level itself) and inter-turn faults from the middle level. An SCIG reaches the same detectability with current transducers alone or with no added sensor. A wound-field machine with a static exciter offers the field current as a further drive-internal channel, whose $2f_e$ ripple detected every inter-winding fault and 80 % of the inter-turn trials at the matching impedance; a brushless exciter would not expose it [30]. Two industrial parameters change these statements: the ratio of current-loop bandwidth to twice the electrical frequency, which for a direct-drive multi-megawatt PMSG at a fundamental near 10 Hz is expected to push the signature further into the controller and the voltage than in these 40–60 Hz machines (a mechanism, not a computed prediction, since the benches' loop gains are undocumented); and the grid-connected, unregulated stator of a doubly-fed generator, whose stator and rotor currents carry the signature that rotor-current and search-coil methods exploit [38], so that whether its machine–drive combination relocates the signature at all is not established by these data.

6.3. Limitations

Each alternative is represented by one physical machine, so the differences reported are differences between three machines; attributing them to topology requires replicated machines or simulated populations validated against the measured units, so the demonstration establishes the method and the mechanisms, not the population-level ranking of the topologies. The fault emulation is resistor-limited and the benches are laboratory scale: a bolted field fault is limited by the leakage reactance of the shorted turns and reaches several times the rated current, so its signature-versus-extent law, and hence every MDE, differs in kind from the one measured here, and the smallest tested fault (2.7–3.8 % of a branch; the data papers give no turn counts) is larger than the one- to three-turn fault that starts an insulation failure in a large winding. The relocation cannot be computed for another drive, whose loop gains, reactances and electrical frequency differ, so Section 6.2 is conditional. Only winding faults and only electrical and controller observ-

ables were studied, at a fixed thermal state and without nuisance transients other than the contactor operation. On the WFSG the fault impedance is confounded with extent, only four tap pairs per class exist at the matching impedance, and there are no healthy trials, so the screen cannot be applied and its transfer entries lack nuisance-transient negatives. Statistically, the pooled null holds its nominal false-alarm rate on average over the envelope, not per speed; the screen rests on nine healthy trials; the fixed feature is chosen on the trials it is evaluated on; the MDE is the median trial of a level and is known only as a bracket on three inter-turn levels; and the required share of 0.9 and the confidence of 0.95 are choices of the requirement. Finally, every MDE describes a short-horizon change detector; a monitor referenced to an old baseline faces a null 1.1 to 4.5 times wider on the key features (Section 5.7).

7. Conclusions

A measure of fault detectability that requires neither a model nor a trained detector has been proposed, the signal-to-drift ratio: a scale-free statistic calibrated on each machine's own healthy non-stationarity, from which a minimum detectable extent, a detectable share and a signature location follow for every combination of machine type, sensor suite and failure-mode class, with a reliability screen, bootstrap intervals and a decision table. Applied to three types of wind-turbine generator with the same tapped winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault and that the faulted phase rides on the nominal extent axis unless sequence quantities are taken as phasor residuals; that where a fault becomes visible depends on the machine–drive combination; that the cheapest sensor suite meeting a requirement depends on the alternative and on the requirement, the middle tested inter-turn level being met by drive-internal quantities on the common bench and the largest on all three machines, and the smallest inter-winding level requiring voltage transducers on the permanent-magnet machine alone; and that a learned detector transfers across the three machines once its features are referenced to healthy behaviour of the target machine. The findings are limited by one physical unit per alternative, laboratory scale, resistor-limited fault emulation and a single failure-mode class; the design-stage instantiation, other failure-mode classes and a cost model in place of the ordinal ranking of suites are the natural extensions.

CRediT authorship contribution statement

Hüseyin Tayyer Canseven: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

Data availability

All data are public under the CC BY 4.0 licence as the InnovaPower benchmark datasets [20–22], archived on Zenodo at https://doi.org/10.5281/zenodo.15741561, https://doi.org/10.5281/zenodo.17161986 and https://doi.org/10.5281/zenodo.13685630.

Acknowledgements

The author thanks the InnovaPower group of the University of São Paulo for publishing the benchmark datasets on which this work rests.

References

[1] M. Riera-Guasp, J. A. Antonino-Daviu, G.-A. Capolino, Advances in electrical machine, power electronic, and drive condition monitoring and fault detection: State of the art, IEEE Trans Ind Electron 62 (2015) 1746–1759. https://doi.org/10.1109/TIE.2014.2375853.

[2] P. J. Tavner, J. Xiang, F. Spinato, Reliability analysis for wind turbines, Wind Energy 10 (2007) 1–18. https://doi.org/10.1002/we.204.

[3] W. T. Thomson, M. Fenger, Current signature analysis to detect induction motor faults, IEEE Ind Appl Mag 7 (2001) 26–34. https://doi.org/10.1109/2943.930988.

[4] G. B. Kliman, W. J. Premerlani, R. A. Koegl, D. Hoeweler, A new approach to on-line turn fault detection in AC motors, in: IAS '96. Conference Record of the 1996 IEEE Industry Applications Conference, volume 1, 1996, pp. 687–693. https://doi.org/10.1109/IAS.1996.557113.

[5] S. B. Lee, R. M. Tallam, T. G. Habetler, A robust, on-line turn-fault detection technique for induction machines based on monitoring the sequence component impedance matrix, IEEE Trans Power Electron 18 (2003) 865–872. https://doi.org/10.1109/TPEL.2003.810848.

[6] S. Nandi, H. A. Toliyat, X. Li, Condition monitoring and fault diagnosis of electrical motors – a review, IEEE Trans Energy Convers 20 (2005) 719–729. https://doi.org/10.1109/TEC.2005.847955.

[7] R. M. Tallam, S. B. Lee, G. C. Stone, G. B. Kliman, J. Yoo, T. G. Habetler, R. G. Harley, A survey of methods for detection of stator-related faults in induction machines, IEEE Trans Ind Appl 43 (2007) 920–933. https://doi.org/10.1109/TIA.2007.900448.

[8] R. M. Tallam, T. G. Habetler, R. G. Harley, Stator winding turn-fault detection for closed-loop induction motor drives, IEEE Trans Ind Appl 39 (2003) 720–724. https://doi.org/10.1109/TIA.2003.811784.

[9] S. M. A. Cruz, A. J. M. Cardoso, Diagnosis of stator inter-turn short circuits in DTC induction motor drives, IEEE Trans Ind Appl 40 (2004) 1349–1360. https://doi.org/10.1109/TIA.2004.834012.

[10] H. Polinder, J. A. Ferreira, B. B. Jensen, A. B. Abrahamsen, K. Atallah, R. A. McMahon, Trends in wind turbine generator systems, IEEE J Emerg Sel Top Power Electron 1 (2013) 174–185. https://doi.org/10.1109/JESTPE.2013.2280428.

[11] W. Yang, P. J. Tavner, C. J. Crabtree, M. Wilkinson, Cost-effective condition monitoring for wind turbines, IEEE Trans Ind Electron 57 (2010) 263–271. https://doi.org/10.1109/TIE.2009.2032202.

[12] W. Yang, P. J. Tavner, C. J. Crabtree, Y. Feng, Y. Qiu, Wind turbine condition monitoring: technical and commercial challenges, Wind Energy 17 (2014) 673–693. https://doi.org/10.1002/we.1508.

[13] M. Basseville, On fault detectability and isolability, Eur J Control 7 (2001) 625–637. https://doi.org/10.3166/ejc.7.625-637.

[14] M. Blanke, M. Kinnaert, J. Lunze, M. Staroswiecki, Diagnosis and Fault-Tolerant Control, 3 ed., Springer, Berlin, Heidelberg, 2016. https://doi.org/10.1007/978-3-662-47943-8.

[15] M. Krysander, E. Frisk, Sensor placement for fault diagnosis, IEEE Trans Syst Man Cybern A Syst Hum 38 (2008) 1398–1410. https://doi.org/10.1109/TSMCA.2008.2003968.

[16] E. Frisk, M. Krysander, J. Åslund, Sensor placement for fault isolation in linear differential-algebraic systems, Automatica 45 (2009) 364–371. https://doi.org/10.1016/j.automatica.2008.08.013.

[17] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, A. K. Nandi, Applications of machine learning to machine fault diagnosis: A review and roadmap, Mech Syst Signal Process 138 (2020) 106587. https://doi.org/10.1016/j.ymssp.2019.106587.

[18] S. J. Pan, Q. Yang, A survey on transfer learning, IEEE Trans Knowl Data Eng 22 (2010) 1345–1359. https://doi.org/10.1109/TKDE.2009.191.

[19] M. Basseville, I. V. Nikiforov, Detection of Abrupt Changes: Theory and Application, Prentice Hall, Englewood Cliffs, NJ, 1993.

[20] R. N. Tominaga, S. S. Barbosa, L. A. Sousa, A. d. S. Lunardi, R. V. Rocha, S. L. Ávila, B. S. Carmo, R. M. Monaro, M. B. d. C. Salles, A benchmark dataset of electrical signals from a permanent magnet synchronous generator for condition monitoring, Data Brief 62 (2025) 112040. https://doi.org/10.1016/j.dib.2025.112040.

[21] R. N. Tominaga, S. S. Barbosa, L. A. Sousa, A. d. S. Lunardi, R. V. Rocha, S. L. Ávila, B. S. Carmo, R. M. Monaro, M. B. d. C. Salles, A condition monitoring dataset based on electrical signals for a squirrel cage induction generator, Data Brief 63 (2025) 112286. https://doi.org/10.1016/j.dib.2025.112286.

[22] R. N. Tominaga, L. A. Sousa, R. V. Rocha, R. M. Monaro, S. L. Ávila, M. B. d. C. Salles, B. S. Carmo, Electrical signals dataset from fixed-speed and variable-speed synchronous generators under healthy and faulty conditions, Data Brief 57 (2024) 111018. https://doi.org/10.1016/j.dib.2024.111018.

[23] J. Coble, J. W. Hines, Identifying suitable degradation parameters for individual-based prognostics, in: Diagnostics and Prognostics of Engineering Systems: Methods and Techniques, IGI Global, Hershey, PA, 2013, pp. 135–150. https://doi.org/10.4018/978-1-4666-2095-7.ch007.

[24] J. A. Hanley, B. J. McNeil, The meaning and use of the area under a receiver operating characteristic (ROC) curve, Radiology 143 (1982) 29–36. https://doi.org/10.1148/radiology.143.1.7063747.

[25] M. Zafarani, E. Bostanci, Y. Qi, T. Goktas, B. Akin, Interturn short-circuit faults in permanent magnet synchronous machines: An extended review and comprehensive analysis, IEEE J Emerg Sel Top Power Electron 6 (2018) 2173–2191. https://doi.org/10.1109/JESTPE.2018.2811538.

[26] M. Arkan, D. K. Perović, P. Unsworth, Online stator fault diagnosis in induction motors, IEE Proc Electr Power Appl 148 (2001) 537–547. https://doi.org/10.1049/ip-epa:20010588.

[27] S. M. A. Cruz, A. J. M. Cardoso, Stator winding fault diagnosis in three-phase synchronous and asynchronous motors, by the extended Park's vector approach, IEEE Trans Ind Appl 37 (2001) 1227–1233. https://doi.org/10.1109/28.952496.

[28] H. Henao, G.-A. Capolino, M. Fernandez-Cabanas, F. Filippetti, C. Bruzzese, E. Strangas, R. Pusca, J. Estima, M. Riera-Guasp, S. Hedayati-Kia, Trends in fault diagnosis for electrical machines: A review of diagnostic techniques, IEEE Ind Electron Mag 8 (2014) 31–42. https://doi.org/10.1109/MIE.2013.2287651.

[29] S. Cheng, P. Zhang, T. G. Habetler, An impedance identification approach to sensitive detection and location of stator turn-to-turn faults in a closed-loop multiple-motor drive, IEEE Trans Ind Electron 58 (2011) 1545–1554. https://doi.org/10.1109/TIE.2010.2064276.

[30] P. Neti, S. Nandi, Stator interturn fault detection of synchronous machines using field current and rotor search-coil voltage signature analysis, IEEE Trans Ind Appl 45 (2009) 911–920. https://doi.org/10.1109/TIA.2009.2018905.

[31] J.-C. Urresty, J.-R. Riba, L. Romeral, Application of the zero-sequence voltage component to detect stator winding inter-turn faults in PMSMs, Electr Power Syst Res 89 (2012) 38–44. https://doi.org/10.1016/j.epsr.2012.02.012.

[32] J. Hang, J. Zhang, M. Cheng, J. Huang, Online interturn fault diagnosis of permanent magnet synchronous machine using zero-sequence components, IEEE Trans Power Electron 30 (2015) 6731–6741. https://doi.org/10.1109/TPEL.2015.2388493.

[33] H. T. Canseven, M. Ercire, M. Cömert, A. Ünsal, N. Sarma, Advanced fault detection of permanent magnet faults in offshore wind turbine generators using finite element analysis and deep transfer learning, Machines 14 (2026) 665. https://doi.org/10.3390/machines14060665.

[34] B. Vaseghi, N. Takorabet, F. Meibody-Tabar, Fault analysis and parameter identification of permanent-magnet motors by the finite-element method, IEEE Trans Magn 45 (2009) 3290–3295. https://doi.org/10.1109/TMAG.2009.2022156.

[35] R. V. Rocha, L. A. Sousa, R. M. Monaro, Experimental platform for evaluation of full-scale variable-speed wind generators under stator winding faults, IEEE Trans Energy Convers 38 (2023) 800–812. https://doi.org/10.1109/TEC.2022.3230260.

[36] E. Gaertner, J. Rinker, L. Sethuraman, F. Zahle, B. Anderson, G. Barter, N. Abbas, F. Meng, P. Bortolotti, W. Skrzypinski, G. Scott, R. Feil, H. Bredmose, K. Dykes, M. Shields, C. Allen, A. Viselli, IEA Wind TCP Task 37: Definition of the IEA 15-Megawatt Offshore Reference Wind Turbine, Technical Report NREL/TP-5000-75698, National Renewable Energy Laboratory, 2020. https://doi.org/10.2172/1603478.

[37] J. H. Friedman, Greedy function approximation: a gradient boosting machine, Ann Stat 29 (2001) 1189–1232. https://doi.org/10.1214/aos/1013203451.

[38] D. Shah, S. Nandi, P. Neti, Stator-interturn-fault detection of doubly fed induction generators using rotor-current and search-coil-voltage signature analysis, IEEE Trans Ind Appl 45 (2009) 1831–1842. https://doi.org/10.1109/TIA.2009.2027406.